Magnetic Resonance Imaging: Variability, Sequences, and Synthetic Reconstruction

Designing a Pipeline for 3D to 2D MRI Acquisition

Unless otherwise noted, all figures were created by researcher

Introduction to MRI

What is MRI?

Magnetic Resonance Images (MRI) are a non-invasive technique to detect diseases, monitor treatment, and examine soft tissues with high-resolution images of organs and tissues

- Uses body's natural magnetic properties
- Machine aligns hydrogen atoms with **magnetic field** and **radiofrequency** pulses to create **detailed** 3D or cross-sectional images
- Measures signal from **hydrogen protons**
- Output = image based on **tissue properties**
- Over 300 types of MRI exams and applications
- **Critical diagnostic** method in nearly every **major area of modern medicine**

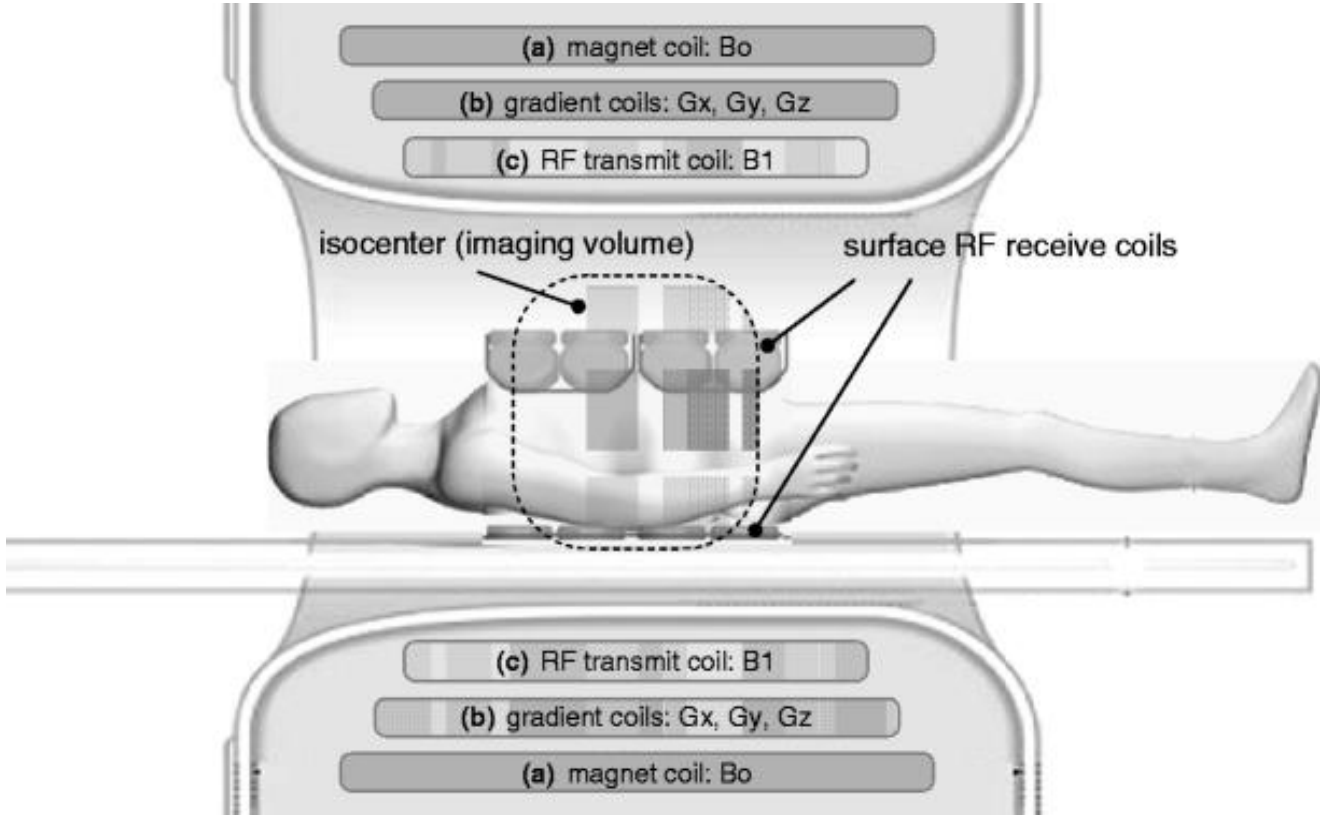
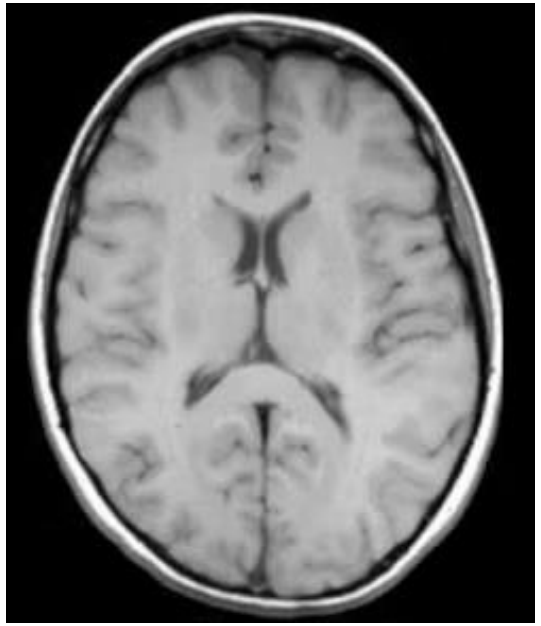


Figure 1. Scheme of an MRI system. *Figure created by Moser et al.*

Common Anatomical Sequences

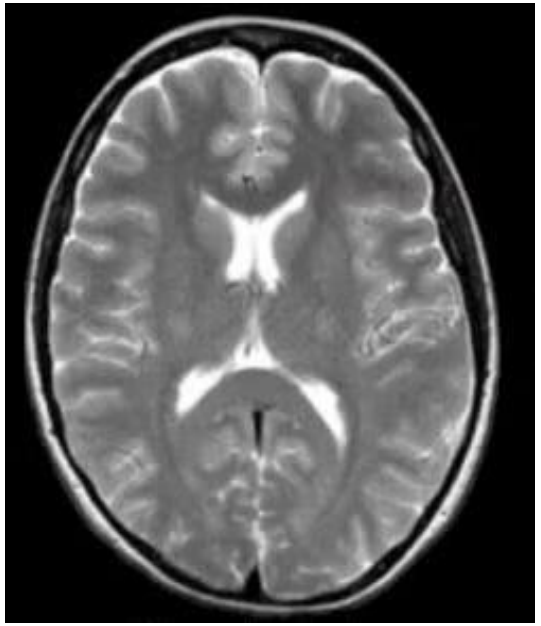
Differences across T1W, T2W, and Flair

T1-weighted



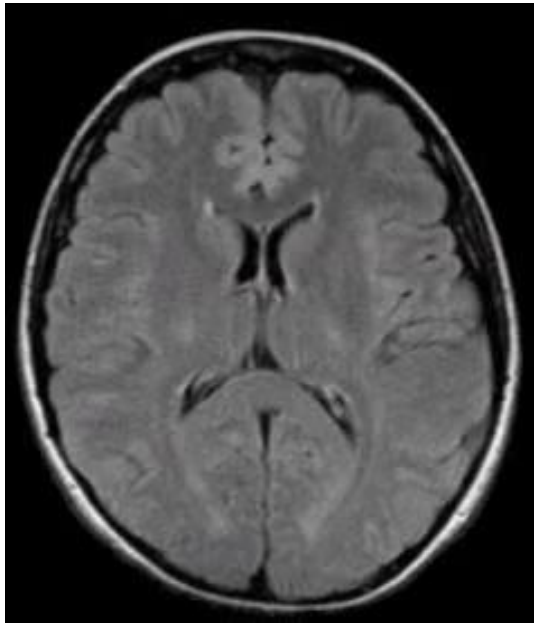
- Water/CSF is dark
- Grey matter darker than white matter
- Best for: anatomy, identifying fat, and post-contrast enhancement

T2-weighted



- Water/CSF bright
- Grey matter brighter than white matter
- Best for: Detecting pathology (edema, inflammation, tumors)

FLAIR

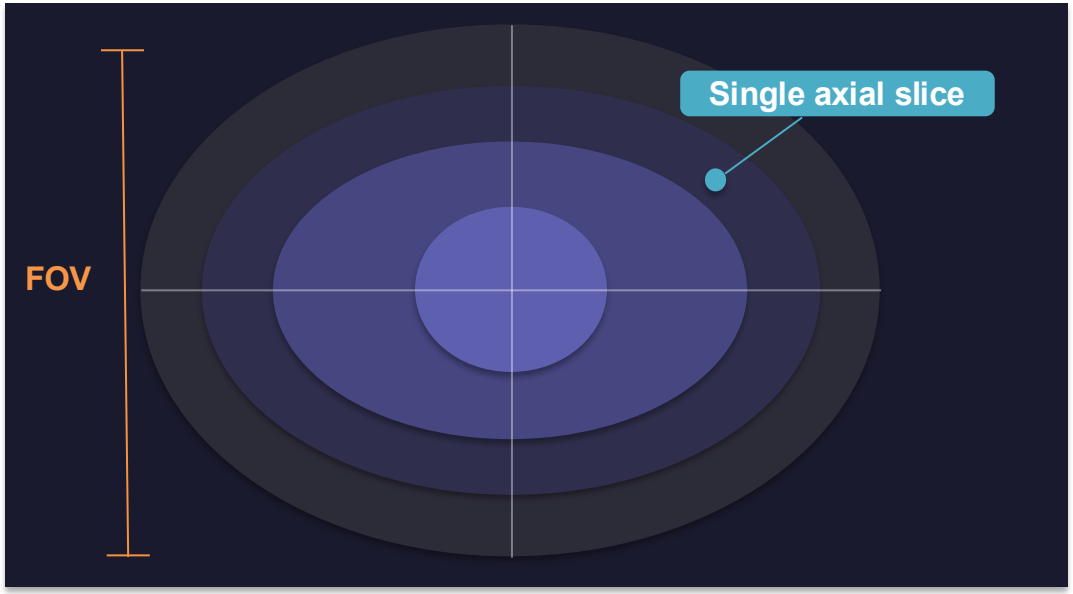


- CSF suppressed (dark)
- Similar contrast to T2, but CSF dark → lesions stand out
- Best for: white matter lesions (edema, MS)

Common Anatomical Sequences

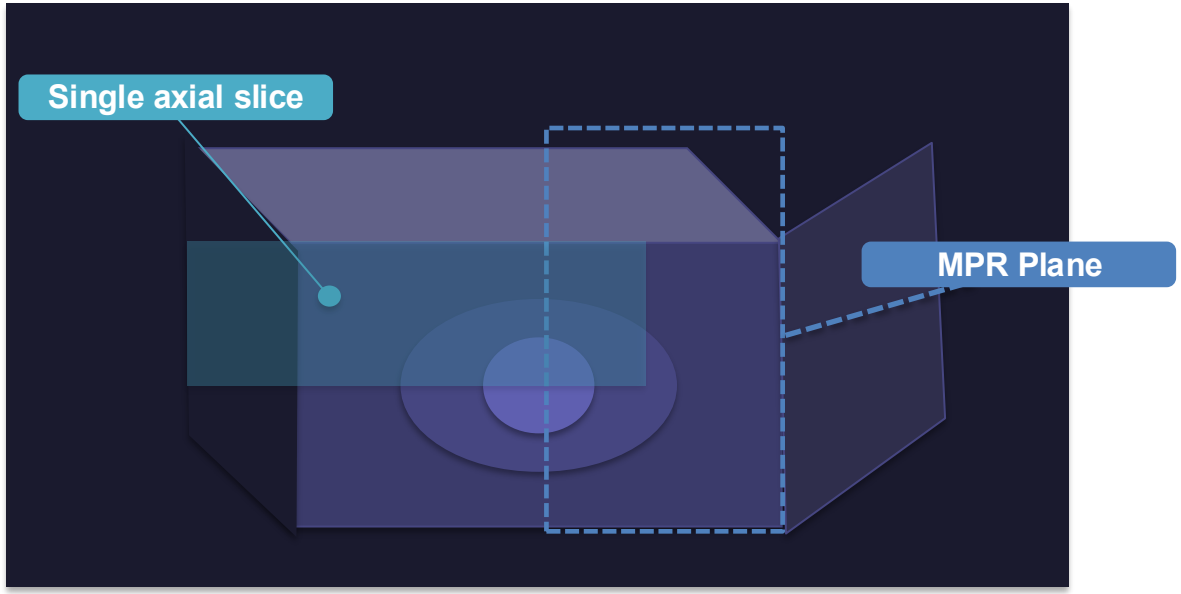
2D vs. 3D readout – acquisition, reconstruction, and clinical tradeoffs

2D Readout (planar acquisition)



Key limitations
Partial volume averaging at boundaries
Cannot reformat post-acquisition
Through-plane resolution limited by SNR
Strengths
Fast time: preferred for dynamic contrast

3D Readout (volumetric acquisition)



Key capabilities
Post-hoc MPR: axial / coronal / sagittal
Volume rendering, MIP, MinIP projections
No interslice gap -> contiguous anatomy
Tradeoffs
Longer time: motion artifact risk higher

Variability

Sources, Harmonization Strategies, and Synthesis

Inherent Sources of MRI Variability

Scanner / vendor
Field inhomogeneity, gradient nonlinearity, coil geometry
GE · Siemens · Philips

Acquisition Protocols
TR/TE, flip angle, bandwidth, matrix size, 2D vs 3D readout
Inter-site protocol drift

Site / longitudinal
Scanner upgrades, head coil changes, subject repositioning
Multi-site studies, aging cohorts

Harmonization and Synthesis Methods - Comparison

Statistical harmonization

ComBat harmonization
Empirical Bayes to remove batch effects
Preserves biological variance · needs site labels

Intensity normalization (N4, WhiteStripe)
Bias field correction + WM peak alignment
Low complexity · does not synthesize contrast

Limitation: cannot generate missing contrasts or simulate unseen scanner configurations

Deep Learning / GAN methods

Contrast synthesis (T1→T2, FLAIR→T1)
Image-to-image translation via pix2pix / CycleGAN
Useful when contrast unavailable at acquisition

Scanner harmonization via GANs
Simulate target-site appearance from source e.g., HACA-GAN, SynSeg-Net, SynthSeg

State of the art: SynthSeg (Billot et al. 2023) —contrast-agnostic segmentation via synthesis

Variability

Overall Conclusion

Head-to-head tradeoffs

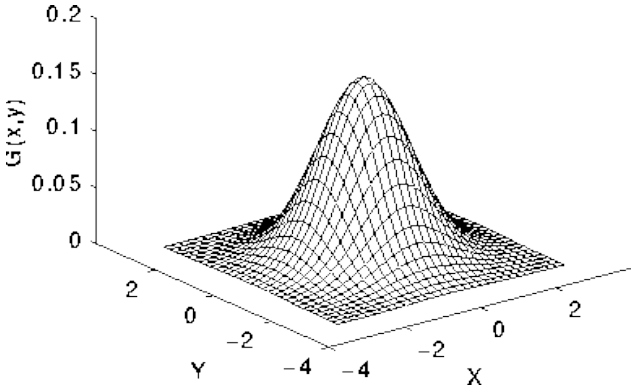
Criterion	Statistical	GL / GAN synthesis
Training data needed	Site labels only	Paired/unpaired image sets
Generates new contrast	No	Yes
Interpretability	High	Low (black box)

Current SOTA favors **learned synthesis** (SynthSeg, contrast-agnostic): **removing** protocol dependency at **inference time**

Physics-Informed Simulation of 2D Flair from 3D Acquisition

Objectives

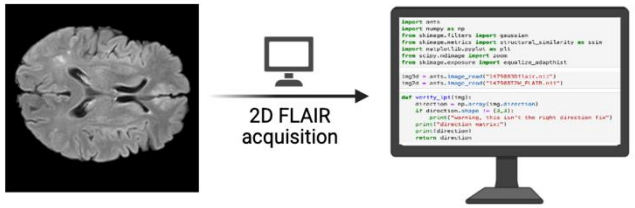
Objective 1



Develop a physics-informed slice profile simulation pipeline

Model 2D FLAIR acquisition from 3D volumes using sinc-Hamming kernels and FWHM-derived Gaussian smoothing

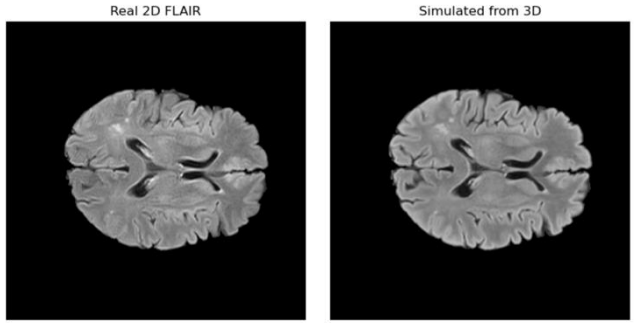
Objective 2



Achieve accurate spatial alignment to clinical 2D acquisition space

Multi-stage registration — rigid initialization followed by SyN deformable refinement, brain-masked to exclude non-tissue regions from the optimization

Objective 3



Quantitatively evaluate simulation fidelity against real 2D FLAIR

Assess structural similarity (SSIM), peak signal-to-noise ratio (PSNR), and normalized cross-correlation (NCC) on brain-masked, normalized image pairs

Physics-Informed Simulation of 2D Flair from 3D Acquisition

Input

1 — ANTsPY · N4ITK

Preprocessing: LPI orientation · N4 bias field correction · Spacing / origin / direction metadata extraction

↓

2 — scipy · skimage

Slice profile simulation: σ computed from target slice thickness (FWHM) · Sinc-Hamming kernel · Gaussian z-smoothing · Best-matching slice found via NCC search

↓

3 — ANTsPY · SyN

Multi-stage registration: Rigid (translation + rotation) initialization · SyN deformable refinement · brain-masked · Resample simulated slice → target voxel grid

↓

4 — ANTsPY · numpy

Post-processing: Histogram matching · Intensity clipping (p99) · CSF suppression correction · Normalization

↓ output

Simulated 2D Flair

Matched to clinical acquisition space

Physics-Informed Simulation of 2D Flair from 3D Acquisition

Quality Evaluation

Key design choice
Physics-based slice profile (Gaussian filtering): interpretable, no training data required

SSIM

Luminance · contrast · brain-masked
Sensitive to perceptual image quality

PSNR

MSE-based fidelity · normalized intensity range
Reference: real 2D FLAIR after masking

NCC

Used for slice selection (best-match z search)
Z-search range: 50-70% of volume extent

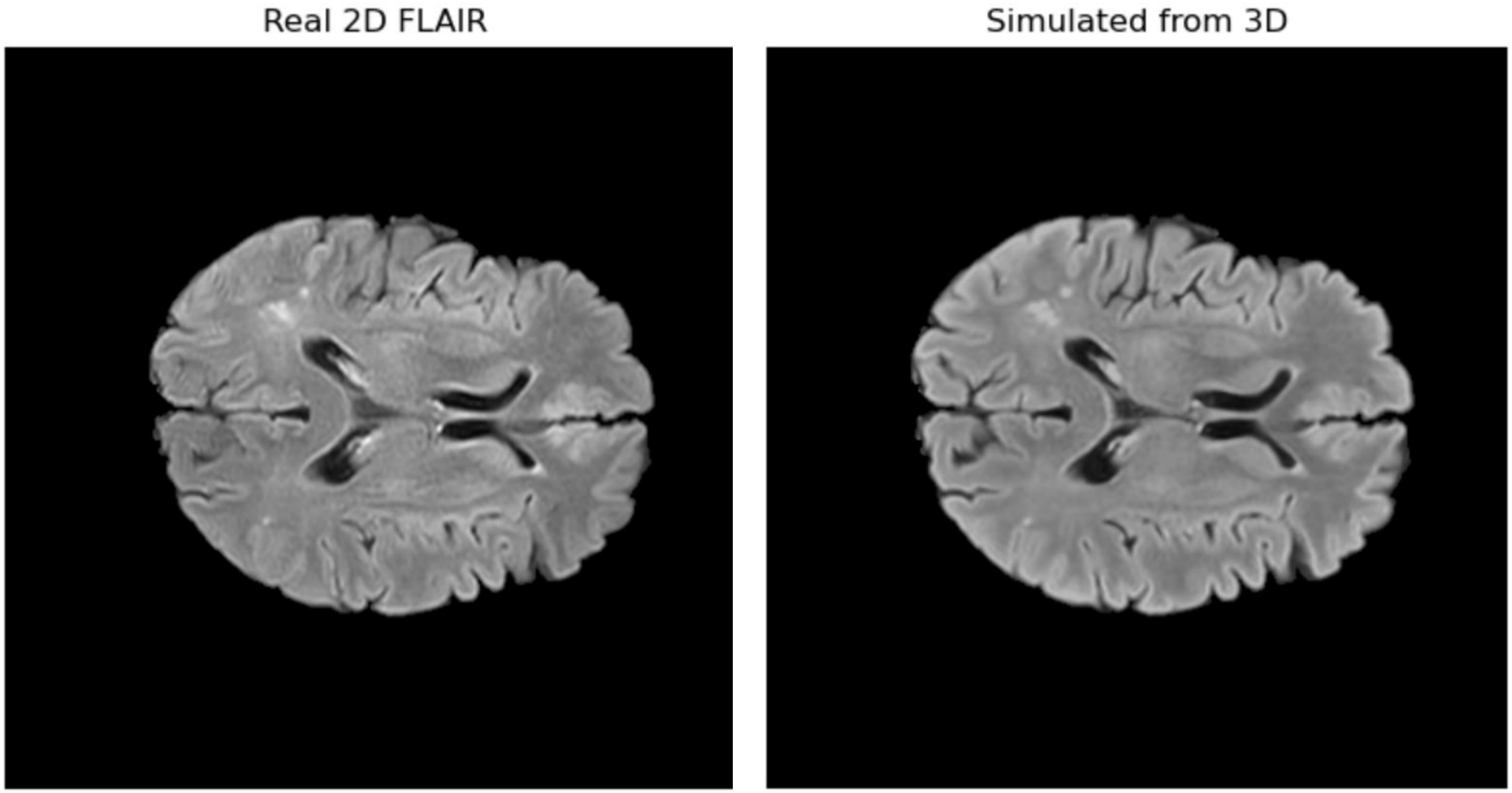
Modality: 3D FLAIR (sub-mm) → simulated 2D T2w-FLAIR

Physics-Informed Simulation of 2D Flair from 3D Acquisition

Preliminary Results

Evaluation

SSIM: 0.9152997085369198
PSNR: 19.138508
NCC: 0.9933788



Future Directions

Extending the physics-informed pipeline toward robust, generalizable simulation

Scanner-specific slice profile: Current kernel assumes idealized Gaussian profile → measure real scanner slice sensitivity functions

Validation & generalization: Validate SSIM/PSNR/NCC across cohort → test generalization, field strength and acquisition protocol

Downstream task evaluation: Evaluate whether simulated 2D improves or matches real 2D on segmentation accuracy

Long-term goal

A generalizable, protocol-agnostic simulation framework given any 3D acquisition, synthesize the equivalent 2D readout for any target scanner, enabling retrospective harmonization of legacy datasets without re-acquisition

Citations

[1] Avants, B.B., Epstein, C.L., Grossman, M., & Gee, J.C. (2008). Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. Medical Image Analysis, 12(1), 26–41. https://doi.org/10.1016/j.media.2007.06.004

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[3] Tustison, N.J., et al. (2021). The ANTsX ecosystem for quantitative biological and medical imaging. Scientific Reports, 11, 9068. https://doi.org/10.1038/s41598-021-87564-6

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